Coffee mug v.s. doughnut

We start with an old joke:



"A topologist is someone who can't tell the different between a coffee cup and a doughnut."

Computational mistakes...

Topologists always make mistakes when doing computations because ...

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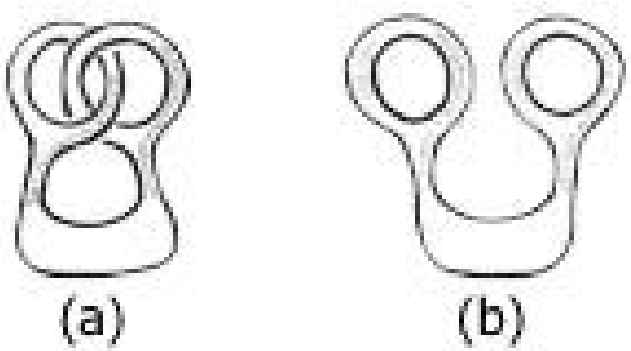
$$1=2 \qquad \text{ONE}=\text{TWO}$$

$$3\neq 4 \qquad \text{THREE}\neq\text{FOUR}$$

in topology!

Unlock two rings

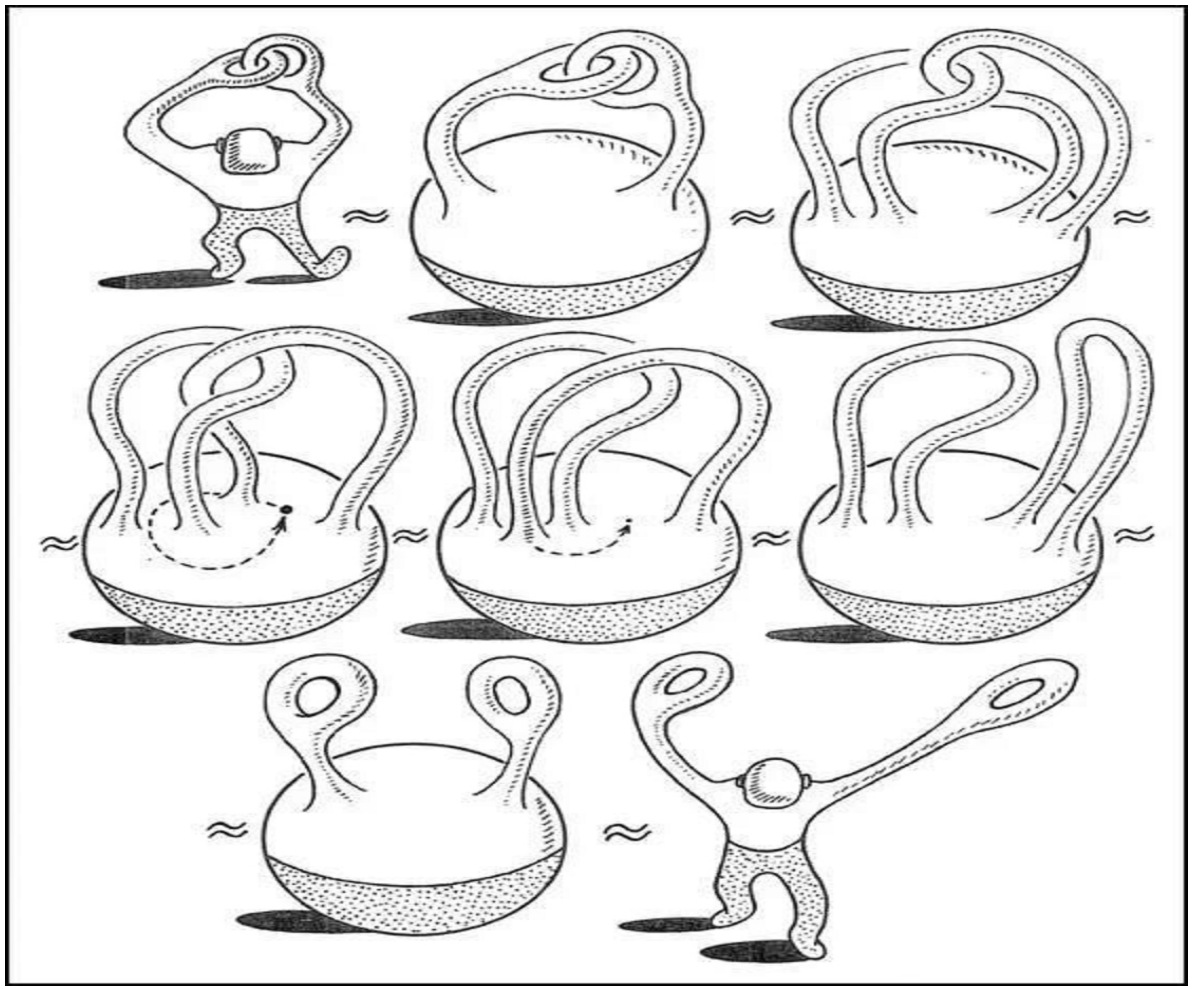
Here is another interesting problem:



Two rings are locked together. Can you unlock the two rings without breaking any of them?

Unlock two rings: The solution

Here is the solution:



Königsberg seven bridge problem

The city of Königsberg[6] consists of four lands. They are connected by seven bridges. The problem is:

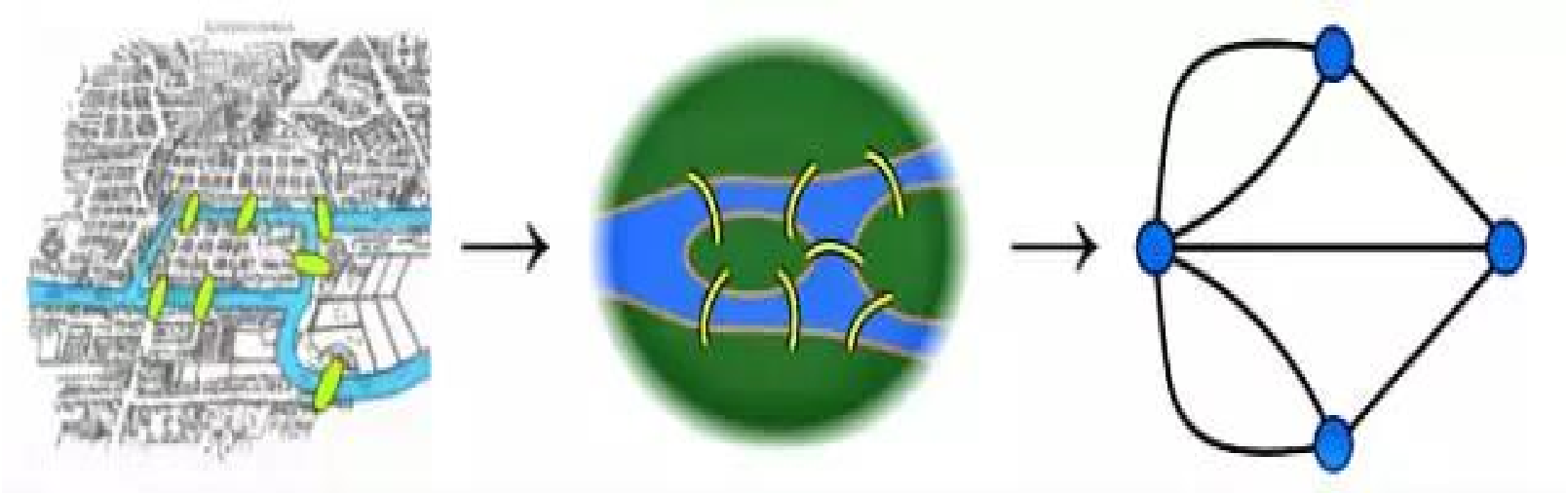
> *Is it possible for a pedestrian to walk across all seven bridges without crossing any bridge twice?*

[6] It is the hometown of the great philosopher I. Kant and the great mathematician D. Hilbert. It was renamed Kaliningrad in 1945.

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Euler's solution to Königsberg seven bridge problem

- The problem did not fit in any existing mathematical framework at that time. It looks like a problem in geometry, but there was no need for distances. Only the relative positions are needed.

[7] By the way, in 1875 the city built a new bridge and the answer became yes!

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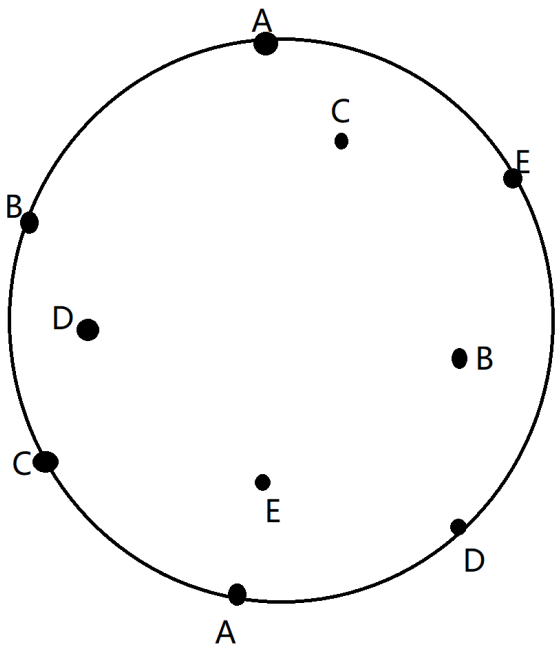
• The birth of topology (and graph theory): Euler solved this problem in 1736 and the answer is no[7]:

- Euler wrote to his friend Ehler in 1736, "*this type of solution bears little relationship to mathematics....*"
- Later in 1736 he wrote to Marinoni, "*it occurred to me to wonder whether it belonged to the geometry of position, which Leibniz had once so much longed for*".
- Finally, still in 1736, he solved this problem and generalize his solution to general case.

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Connecting points

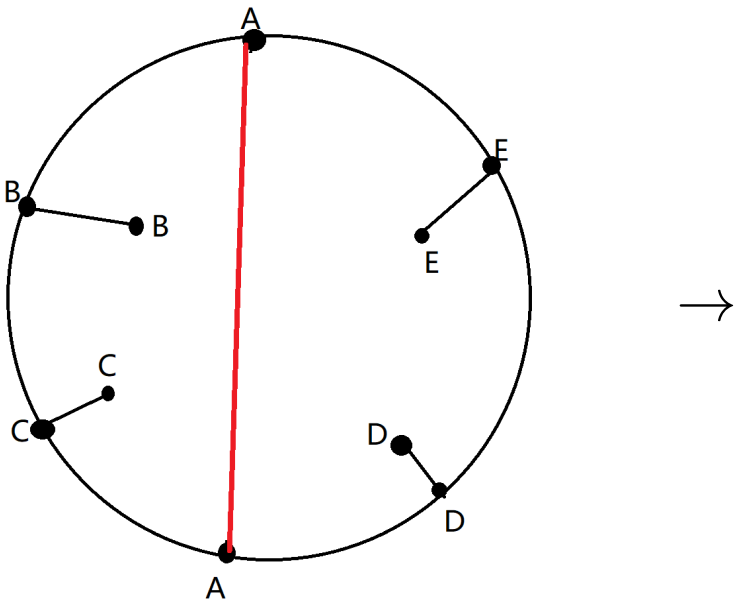
Here is another example that you can apply the idea of topology in solving problems:



Question: Can you connect points marking the same letter by non-intersecting curves inside the circle?

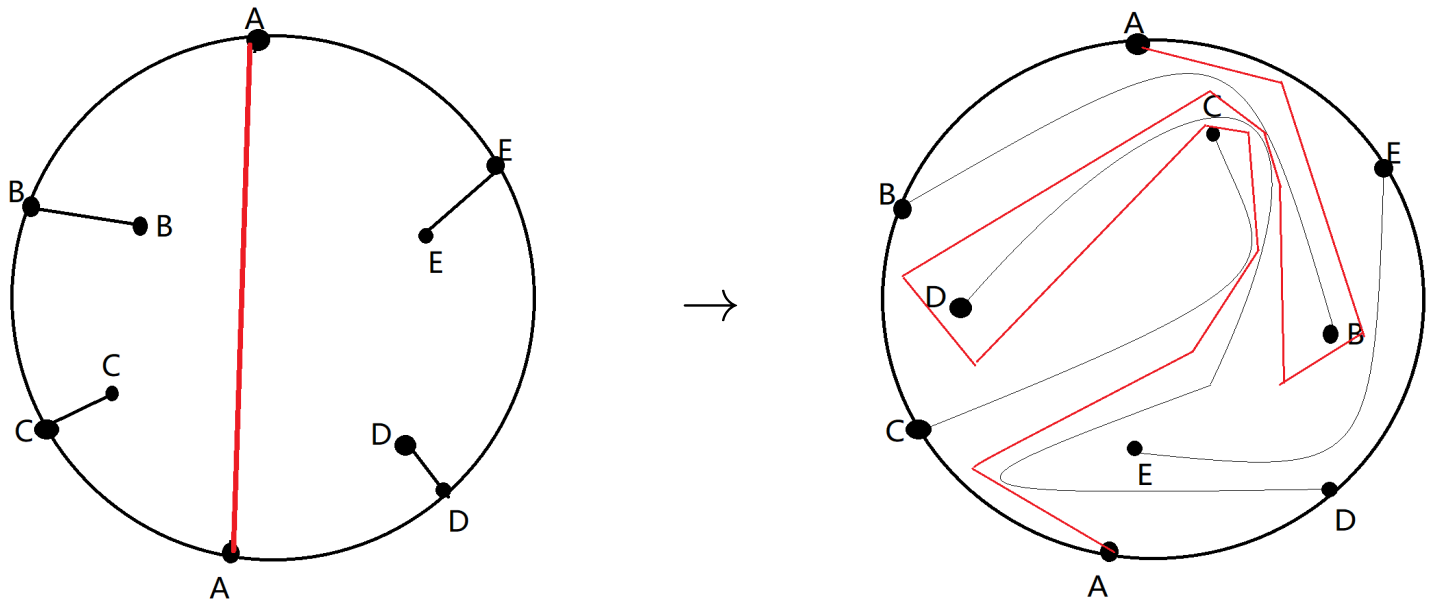
Connecting points: Solution

To solve this problem "topologically", you first continuously move the points inside the circle to positions that you can easily solve the problem. Then continuously move the points back and stretch the curves you have drawn so that they are always non-intersecting:



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What's next ☞

1. What is topology...
   - A: The origin of the subject
   - B: Some abstract words
   - C: Some concrete pictures
2. Some topological theorems that we have learned/will learn
   - Part I: Point-Set Topology and Applications to Analysis
   - Part II: Algebraic/Geometric Topology

As we mentioned above, topology structure is one of the mother structures in mathematics. In fact, we have used topology in many places in earlier courses. Here we list some of them, together with some related theorems that we might cover later in this course.

Part I: Point-Set Topology and Applications to Analysis

A: The Intermediate Value Theorem

We start with a baby theorem:

Baby-Theorem

Suppose you are given a point A in the upper half plane, and a point B in the lower half plane. If you draw a planar curve connecting A to B, the curve must intersect the x-axis.

A: The Intermediate Value Theorem

• At the very beginning of mathematical analysis, we learned following more general theorem, which is one of the most important theorems in mathematics:

Theorem (The Intermediate Value Theorem)

Let $f : [a, b] \to \mathbb{R}$ be a continuous function. Then for any value y between $f(a)$ and $f(b)$, there exists $c \in [a, b]$ s.t. $f(c) = y$.

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- Topological property behind the Intermediate Value Theorem:

the *connectedness* of $[a, b]$.

A: The Intermediate Value Theorem

In this course we will prove:

Theorem (The Generalized Intermediate Value Theorem)

Let X, Y be topological spaces, and let $f : X \to Y$ be a continuous map. Then for any connected subset $A \subset X$, the image $f(A)$ is a connected subset in Y.

A: The Intermediate Value Theorem

Here is a simple but beautiful application of IVT:

Proposition

Let $\mathcal{C}$ be a closed planar curve. Then there exists a planar square that contains the curve $\mathcal{C}$, so that the curve touches all four sides (vertices count as points in both sides) of the square.

